2.1 Using Fundamental Identities

Name _____ Period ____ Date _____

Pre-Calculus 2.1 · Student sheet

I can…

- recognize and write the fundamental trigonometric identities.
- use the fundamental identities to evaluate trig functions.
- use the fundamental identities to simplify, factor, and add trig expressions.

Vocabulary · fill in the missing word

Reciprocal identities — $\csc u =$ _____, $\sec u = 1/\cos u$, $\cot u = 1/\tan u$

Quotient identities — $\tan u =$ _____, $\cot u = \cos u/\sin u$

Pythagorean identities — $\sin^2 u + \cos^2 u =$ _____, $1 + \tan^2 u = \sec^2 u$, $1 + \cot^2 u = \csc^2 u$

Cofunction identities — $\sin(\pi/2 - u) =$ _____, and so on

Even/odd identities — $\cos(-u) = \cos u$ (even); sin, tan, csc, cot of $-u = -$(the _____) (odd)

Part 1 · Identity discovery

Pick your own angle θ (not 30°, 45°, 60°, or 90°): $\theta =$ _____. Use a calculator in degree mode. Round to four decimal places.

Expression	Value	Compare with	Value
$\sin^2\theta + \cos^2\theta$		—	
$1 + \tan^2\theta$		$1/\cos^2\theta$	
$\sin(-\theta)$		$-\sin\theta$	
$\cos(-\theta)$		$\cos\theta$	
$\cos(90° - \theta)$		$\sin\theta$	

Compare with a pair that used a different angle. Write each pattern as an equation:

Part 2 · Notes: fill in the blanks

Reciprocal: $\csc u =$ _____ $\sec u =$ _____ $\cot u =$ _____

Quotient: $\tan u =$ _____ $\cot u =$ _____

Pythagorean: $\sin^2 u + \cos^2 u =$ _____ $1 + \tan^2 u =$ _____ $1 + \cot^2 u =$ _____

Cofunction: $\sin(\pi/2 - u) =$ _____ $\tan(\pi/2 - u) =$ _____

Even/odd: $\cos(-u) =$ _____ $\sin(-u) =$ _____ $\tan(-u) =$ _____

Part 3 · Examples (work them with the class)

Example 1

Use the given values to evaluate (if possible) all six trigonometric functions.

$\sec u = -\frac{3}{2}$ and $\tan u > 0$

Example 2

Simplify $\sin x \cos^2 x - \sin x$.

Example 3

Factor $4\tan^2\theta + \tan\theta - 3$.

Example 4

Factor $\csc^2 x - \cot x - 3$.

Example 5

Perform the addition and simplify.

$\sin \frac{\theta}{1 + \cos\theta} + \cos\theta/\sin\theta$

Example 6

Rewrite $\frac{1}{1 + \sin x}$ so that it is not in fractional form.

Part 4 · Partner practice

Partner 1

Simplify $\cot\theta \sec\theta$.

Partner 2

Given $\sin\theta = -\frac{3}{5}$ and $\cos\theta > 0$, find $\tan\theta$ and $\sec\theta$.